Computational Engineering & Data Analysis

MECH2700

Write the symbolic form first · keep units · check assumptions

ERROR & CONVERGENCE

$E_a = |x - \hat{x}| \cdot E_r = \frac{E_a}{|x|} \cdot E_\% = 100E_r$
$E(h) \approx Ch^p$, so $p \approx \frac{\ln\left(\frac{E_h}{E_{\frac{h}{2}}}\right)}{\ln 2}$
Halving h reduces error by 2^p
Stop when $|x_{k+1} - x_k| \le \varepsilon_a$ or $\frac{|x_{k+1}-x_k|}{|x_{k+1}|} \le \varepsilon_r$

INTERPOLATION

$p_{n(x)} = \sum_{i=0}^{n} y_i L_{i(x)}$
$L_{i(x)} = \prod_{j=0, j\neq i} \frac{x - x_j}{x_i - x_j}$
$f(x) - p_{n(x)} = f^{(n+1)}(\xi) \frac{1}{(n+1)!} \prod_{i=0}^{n(x-x_i)}$
Newton form: $p_{n(x)} = a_0 + a_1(x - x_0) + \ldots + a_n \prod_{j=0}^{n-1}(x - x_j)$
$a_k = f[x_0, \ldots, x_k]$, with $f[x_i, x_j] = \frac{f(x_j) - f(x_i)}{x_j - x_i}$

CHEBYSHEV INTERPOLATION

$T_0 = 1, T_1 = x, T_{n+1} = 2xT_n - T_{n-1}$
Roots on $[-1, 1]$: $t_k = \cos\left((2k-1)\frac{\pi}{2n}\right)$
Map to $[a, b]$: $x_k = \frac{a+b}{2} + (b-a)\frac{t_k}{2}$
Use Chebyshev nodes to reduce endpoint oscillation

FINITE DIFFERENCES

$f'(x_i) \approx \frac{f_{i+1} - f_i}{h}$ forward, $O(h)$
$f'(x_i) \approx \frac{f_i - f_{i-1}}{h}$ backward, $O(h)$
$f'(x_i) \approx \frac{f_{i+1} - f_{i-1}}{2h}$ central, $O(h^2)$
$f''(x_i) \approx \frac{f_{i+2} - 2f_{i+1} + f_{i-1}}{h^2}$, $O(h^2)$
Boundary $O(h^2)$: $f'(x_0) \approx \frac{-3f_0 + 4f_1 - f_2}{2h}$

NEWTON–COTES

$T_1 = (b-a)\frac{f(a)+f(b)}{2}$
Composite trapezoid, $h = \frac{b-a}{n}$: $T_n = h\left[\frac{f_0}{2} + \sum_{i=1}^{n-1} f_i + \frac{f_n}{2}\right]$
Composite Simpson $\frac{1}{3}$ (n even): $S_n = \frac{h}{3}\Big[f_0 + f_n + 4\sum_{i \text{ odd}} f_i + 2\sum_{i \text{ even}, i\neq 0, n} f_i\Big]$
Simpson $\frac{3}{8}$ (n multiple of 3): weights $1, 3, 3, 2, \ldots, 3, 3, 1$ and factor $3\frac{h}{8}$

GAUSSIAN QUADRATURE

$\int_a^b f(x)\,\mathrm{d}x = \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{a+b}{2} + (b-a)\frac{t}{2}\right)\mathrm{d}t$
$\approx \frac{b-a}{2}\sum_i w_i f(x(t_i))$
2-point: $t = \pm\frac{1}{\sqrt{3}}$, $w = (1, 1)$
3-point: $t = \left(-\sqrt{\frac{3}{5}}, 0, \sqrt{\frac{3}{5}}\right)$, $w = \left(\frac{5}{9}, \frac{8}{9}, \frac{5}{9}\right)$
n Gauss points are exact through degree $2n - 1$

DIRECT LINEAR SYSTEMS

$Ax = b$ · residual $r = b - Ax$
Elimination: $m_{ik} = \frac{a_{ik}}{a_{kk}}$; row $i \to$ row $i - m_{ik}$ row k
$A = LU$; solve $Ly = b$, then $Ux = y$
Forward: $y_i = \frac{b_i - \sum_{j<i} l_{ij} y_j}{l_{ii}}$
Back: $x_i = \frac{y_i - \sum_{j>i} u_{ij} x_j}{u_{ii}}$
Partial pivot on largest available $|a_{ik}|$ before division

NORMS & CONDITIONING

$\|x\|_1 = \sum_i |x_i| \cdot \|x\|_2 = \sqrt{\sum_i x_i^2} \cdot \|x\|_\infty = \max_i |x_i|$
$\|A\|_\infty = \max_i \sum_j |a_{ij}|$
$\kappa(A) = \|A\|\|A^{-1}\|$
$\frac{\|\delta x\|}{\|x\|} \le \kappa(A)\frac{\|\delta b\|}{\|b\|}$ approximately

ITERATIVE LINEAR SYSTEMS

$x_i^{(k+1)} = \frac{b_i - \sum_{j\neq i} a_{ij} x_j^*}{a_{ii}}$
Jacobi: every $x_j^* = x_j^{(k)}$
Gauss–Seidel: use new values for $j < i$, old for $j > i$
Relaxation: $x^{(k+1)} = (1-\omega)x^{(k)} + \omega x_{\text{GS}}^{(k+1)}$, usually $0 < \omega < 2$
Sufficient convergence: $|a_{ii}| > \sum_{j\neq i} |a_{ij}|$

LEAST SQUARES

$S = \sum_{i(y_i - f(x_i))}^{2}$
For $f(x) = \sum_{j=0}^{m} a_j \varphi_{j(x)}$: $Aa \approx y$
Normal equations: $A^T Aa = A^T y$
Line $y = a + bx$: $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$
$a = |y| - b|x|$

BISECTION & FIXED POINT

$m_k = \frac{a_k + b_k}{2}$; retain the half with a sign change
After n bisections: width $= \frac{b-a}{2^n}$
$n \ge \left\lceil \log_2\left(\frac{b-a}{\varepsilon}\right) \right\rceil$ for width $\le \varepsilon$
Fixed point: $x_{k+1} = g(x_k)$; local convergence when $|g'(r)| < 1$

NEWTON & SECANT ROOTS

$x_{k+1} = x_k - \frac{f(x_k)}{f'}(x_k)$ (quadratic near a simple root)
Secant: $x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$
Check bracket/sign, derivative size, residual and step; do not divide by a near-zero denominator

GOLDEN-SECTION SEARCH

$\tau = \frac{\sqrt{5}-1}{2} \approx 0.618$
$x_1 = b - \tau(b-a)$, $x_2 = a + \tau(b-a)$
Discard the side with the larger value for minimisation; reuse one function value
Interval width after n: $L_n = \tau^n L_0$
Requires a unimodal objective on the bracket

ODE SETUP & EULER

$y' = f(t, y)$, $y(t_0) = y_0$, $t_{n+1} = t_n + h$
Euler: $y_{n+1} = y_n + hf(t_n, y_n)$; global $O(h)$
Second-order $y'' = g(t, y, y')$: set $z_1 = y$, $z_2 = y'$; then $z_{1'} = z_2$, $z_{2'} = g(t, z_1, z_2)$

RK2 & RK4

$k_1 = f(t_n, y_n), k_2 = f(t_n + \alpha h, y_n + \alpha h k_1)$
$y_{n+1} = y_n + h\left[\left(1 - \frac{1}{2\alpha}\right)k_1 + \frac{k_2}{2\alpha}\right]$
$\alpha = 1$ gives Heun; $\alpha = \frac{1}{2}$ gives midpoint
RK4: $k_1 = f(t, y)$; $k_2 = f\left(t + \frac{h}{2}, y + h\frac{k_1}{2}\right)$; $k_3 = f\left(t + \frac{h}{2}, y + h\frac{k_2}{2}\right)$; $k_4 = f(t + h, y + hk_3)$
$y_{n+1} = y_n + \frac{h(k_1 + 2k_2 + 2k_3 + k_4)}{6}$; global $O(h^4)$

GRID DERIVATIVES

$x_i = i\Delta x$, $t^n = n\Delta t$, $u_i^n \approx u(x_i, t^n)$
$u_t \approx \frac{u_i^{n+1} - u_i^n}{t}$
$u_x \approx \frac{u_i^n - u_{i-1}^n}{x}$ for right-going upwind
$u_{xx} \approx \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta} x^2$

DIFFUSION / HEAT

$u_t = Du_{xx}$; $\sigma = D\Delta\frac{t}{\Delta}x^2$
Explicit: $u_i^{n+1} = \sigma u_{i-1}^n + (1 - 2\sigma)u_i^n + \sigma u_{i+1}^n$
1D stability: $0 \le \sigma \le \frac{1}{2}$
Implicit: $-\sigma u_{i-1}^{n+1} + (1 + 2\sigma)u_i^{n+1} - \sigma u_{i+1}^{n+1} = u_i^n$

CONVECTION & WAVES

$u_t + vu_x = 0$
For $v > 0$: $u_i^{n+1} = u_i^n - C(u_i^n - u_{i-1}^n)$
For $v < 0$: $u_i^{n+1} = u_i^n - C(u_{i+1}^n - u_i^n)$
$C = v\Delta\frac{t}{\Delta}x$, $|C| \le 1$
$u_{tt} = c^2 u_{xx}$: $u_i^{n+1} = 2u_i^n - u_i^{n-1} + C^2(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$
2D adds $C_x^2\left(u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n\right) + C_y^2\left(u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n\right)$
1D CFL $|C| \le 1$; 2D requires $C_x^2 + C_y^2 \le 1$

ELLIPTIC / LAPLACE

$u_{xx} + u_{yy} = 0$
Equal grid spacing: $u_{i,j} = \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}}{4}$
Matrix row: centre -4, each interior neighbour $+1$; move known boundary terms to the RHS
Dirichlet gives u; Neumann gives a derivative and needs a one-sided/ghost-node relation

EXAM METHOD

- Write the continuous problem, then the discrete formula
- Keep old/new time levels separate; copy arrays before explicit updates
- Apply boundary values outside the interior loop
- State accuracy, convergence or CFL condition
- Verify with residual, bracket, units, symmetry or a refined step
- If the answer is unstable, first check sign, h powers and overwritten values